

ECON100B - SECTION 19

Monetary Policy and Phillips Curve

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ANNOUNCEMENTS/AGENDA

Announcements

- PSET 3 due on Friday, 4/17

Agenda

- Monetary Policy and the Phillips Curve

Question of the Day

Suppose I have an economy that starts out with an $\bar{a} = 0$, $\bar{b} = 0.6$, $R_t = 0.04$, and $\bar{r} = 0.04$. Graph this initial economy. What should my short run output be in this scenario?

Say my economy has had a long-term shock that has led to a new MPK of 0.05. Along with this MPK shock, I am now more sensitive to changes in the real interest rate, such that my $\bar{b} = 0.8$. My real interest rate (for now) stays constant at 0.04. Graph this new economy in relation to the original economy. What should my short run output be in this scenario?

Question of the Day Answer

In our original economy, we have a short run output of 0 and downward sloping IS curve.

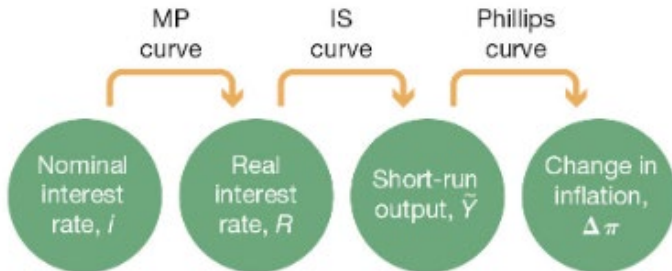
In the new economy, we have a short run output of 0.8%, and our IS curve shifts outwards and becomes flatter.

Monetary Policy and the Phillips Curve



The Central Bank and Monetary Policy

- The main policy tool used by the U.S. Federal Reserve is the federal funds rate, which is the interest rate paid from one bank to another for overnight loans

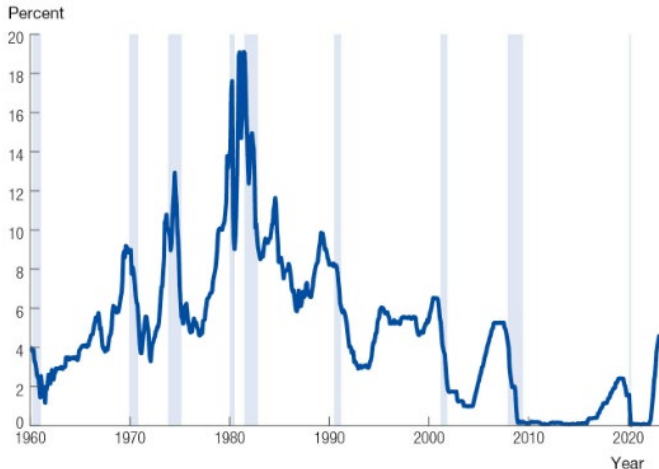


How does the Federal Funds Rate Work?

- Banks and financial institutions routinely lend to and borrow from one another daily for several reasons
- The Central Bank is able to set this rate by stating that they are willing to borrow or lend any amount at a specific rate. In the U.S., this is the federal funds rate
- Every bank will charge this same rate on overnight loans, otherwise banks would just borrow at the lower rate from the central bank
- The Central Bank's willingness to borrow at this same rate helps to disincentivize any bank from charging a lower overnight loan rate

The Federal Funds Rate over Time

The Federal Funds Rate



Recalling the Fisher Equation

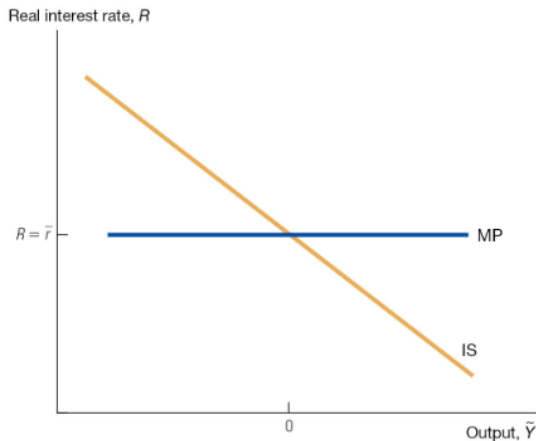
$$i_t = R_t + \pi_t.$$

$$R_t = i_t - \pi_t.$$

- Changes in the nominal interest rate will lead to changes in the real interest rate as long as they are not offset by corresponding changes in inflation
- The sticky inflation assumption states that inflation takes time to change and will not do so immediately
- Practically, what this means is that central banks set the real interest rate in the short run

The IS/MP Diagram

The MP Curve in the IS/MP Diagram



The Phillips curve, revisited

Economy-wide inflation will be equal to what firm owners across the economy expect inflation will be, plus/minus the effect of any booms or busts

$$\pi_t = \underbrace{\pi_t^e}_{\text{expected inflation}} + \underbrace{\bar{v}\tilde{Y}_t}_{\text{demand conditions}} .$$

with the assumption:

$$\pi_t^e = \pi_{t-1}.$$

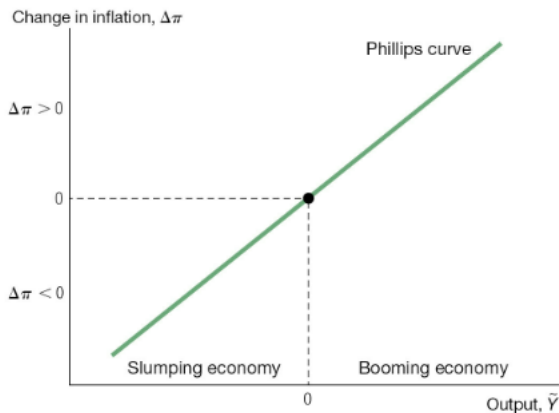
So we can rewrite our Phillips Curve

$$\pi_t = \pi_{t-1} + \bar{v}\tilde{Y}_t.$$

$$\Delta\pi_t = \bar{v}\tilde{Y}_t.$$

The Phillips curve

The Phillips Curve



Unexpected Shocks

- Our model of inflation currently considers previous inflation/expected inflation and the state of the economy
- But what about sudden price shocks? For example, the current conflict in the Middle East?
- We introduce a shock term to inflation as well:

$$\pi_t = \pi_{t-1} + \bar{v}\tilde{Y}_t + \bar{o}.$$

$$\Delta\pi_t = \bar{v}\tilde{Y}_t + \bar{o}.$$

- This is the final form of the Phillips curve which we'll use

Types of Inflation

- Cost-push inflation: Inflation that is caused because the costs of inputs push the costs of goods and services up
- Demand-pull inflation: Changes in the aggregate demand in the economy affect the inflation rate

Phillips Curve Practice Problems

1. If we observe a shock in oil prices, what happens to the movement of our Phillips Curve compared to baseline?
2. Under the Phillips Curve framework, what component of the Phillips Curve is considered cost-push inflation, and which component is considered demand-pull inflation?
3. We are told that inflation in the previous period was 4%, and our short run output is expected to be -2% with a sensitivity parameter of 0.5. We are also hit with a sudden supply chain shock that has increased a 1.5% shock to inflation. What is our final inflation rate in this current period? How would we show this on a graph with the Phillips Curve?

Phillips Curve Practice Solutions

1. Our Phillips Curve pushes left and out: we expect to see higher levels of inflation for any short-run output levels because of the oil price shock.
2. Shocks or “o” are considered cost-push inflation and short-run output effects are considered demand-pull inflation
3. We’d expect a 0.5% inflation rate. Our curve pushes left-ward and outward, but we also move down the new curve.

Modeling the 1970s Inflation Crisis using the Complete Short-Run Model

See lecture slides!

Putting it all together: IS-MP-PC Model

3. (36 points total) *Thanks a lot, Generation X, You Broke the Short Run.*

“Generation X” is the term for Americans born roughly between the 1960s and 1980s. People born later are commonly called “Millennials.” Recall the standard IS, MP, and PC Curves we saw in class:

$$\tilde{Y} = \bar{a} - \bar{b}(R - \bar{r}) \quad (4)$$

$$R = R^{ff} + \bar{p} \quad (5)$$

$$\Delta\pi = \bar{v}\tilde{Y} + \bar{o} \quad (6)$$

where R is the real interest rate paid by borrowers; $\bar{r}$ is the marginal product of capital; R^{ff} is the real federal funds rate; $\bar{p}$ is the risk premium charged by private lenders; π is the inflation rate; and $\bar{o}$ is an external inflation shock.

Imagine that it is 2002, and the economy has emerged from the recession of 2001. Generation X'ers feel optimistic about the future, and they increase spending on housing as a share of permanent income, racking up debt.

- (a) (6 points) Illustrate this development *and any response by the Federal Reserve* in the IS-MP and Phillips Curve diagrams below. Label and describe what you've drawn. *If a curve shifts twice in this answer, show that.*

Putting it all together: IS-MP-PC Model

Now imagine that it is 2006, and a major crisis hits the housing market. Generation X'ers discover that their incomes are not sufficient to make their mortgage payments on the enormous houses they starting buying in 2002.

Homeowners sharply reduce their consumption spending in hopes of avoiding default on their mortgages. But it's too little, too late.

- (b) (6 points) Illustrate this development *and any response by the Federal Reserve* in the IS-MP and Phillips Curve diagrams below. Label and describe what you've drawn. *If a curve shifts twice in this answer, show that.* [You don't need to show the events from part (a).]